Resilience Patterns for HTTP Clients

This guide covers the retry and backoff conventions used by internal services when calling downstream APIs.

All outbound HTTP calls must use exponential backoff with jitter. The base delay is 1 second, doubling on each attempt, capped at 30 seconds, for a maximum of 5 attempts before the call is considered failed.

Only retry on transient failures: HTTP 429, 502, 503, 504, and connection timeouts. Do not retry on 4xx client errors other than 429, since retrying a malformed request will never succeed.

Jitter is applied by multiplying the computed delay by a random factor between 0.8 and 1.2. This prevents thundering-herd retry storms when many clients fail at the same time.

Circuit breakers sit in front of the retry layer. After 5 consecutive failures to a given host, the breaker opens for 60 seconds and short-circuits further calls immediately, returning a fast failure instead of waiting for a timeout.